



RS003 Week 4 — English Worksheet

Topic: A Three-Part Base — Ground + Base + Pivot · **Course:** Pygame Turret · **Time:** about 45 minutes

This week your base grows from one rectangle into a **three-part scene**: a long ground bar across the bottom, the base body on top of it, and a small pivot circle where the gun will spin. You also learn that **draw order** decides what sits in front.

Words to know: **ground**, **palette**, **outline (border)**, **draw order**, **pivot**

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
pygame.draw.rect(screen, EARTH, (0, HEIGHT - 50, WIDTH, 50))
```

This rectangle starts at the far left and is as wide as the whole screen. What does it look like?


```
pygame.draw.rect(screen, DARK_GREEN, (CENTER_X - 35, HEIGHT - 90, 70, 45))
pygame.draw.rect(screen, BLACK, (CENTER_X - 35, HEIGHT - 90, 70, 45), 2)
```

The same rectangle is drawn twice. The second one has an extra **2** at the end. What does that **2** add?


```
screen.fill(SKY)
pygame.draw.rect(screen, EARTH, ...) # ground
pygame.draw.circle(screen, SILVER, ...) # pivot
```

Things are drawn from first to last. Which one ends up on top — the sky, the ground, or the pivot?

2 · Run It 🏃

🎯 Type the example, run it, and name the three parts

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

SKY = (135, 206, 235)
EARTH = (95, 60, 30)
DARK_GREEN = (0, 100, 0)
LIGHT_GREEN = (50, 150, 50)
BLACK = (0, 0, 0)
SILVER = (180, 180, 180)

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    screen.fill(SKY)
    pygame.draw.rect(screen, EARTH, (0, HEIGHT - 50, WIDTH, 50))
    pygame.draw.rect(screen, LIGHT_GREEN, (0, HEIGHT - 55, WIDTH, 8))
    pygame.draw.rect(screen, DARK_GREEN, (CENTER_X - 35, HEIGHT - 90, 70, 45))
    pygame.draw.rect(screen, BLACK, (CENTER_X - 35, HEIGHT - 90, 70, 45), 2)
    pygame.draw.circle(screen, SILVER, (CENTER_X, HEIGHT - 90), 12)
    pygame.draw.circle(screen, BLACK, (CENTER_X, HEIGHT - 90), 12, 2)
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

Name the three parts you can see and where each one is:

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — The ground should appear **behind** the base. Right now the ground is drawn last and covers the base.

```
pygame.draw.rect(screen, DARK_GREEN, (CENTER_X - 35, HEIGHT - 90, 70, 45))
pygame.draw.rect(screen, EARTH, (0, HEIGHT - 50, WIDTH, 50))
```

Hint: whatever is drawn last sits on top.

Write the fixed code (show both lines in the right order):

Why was it wrong? Why does your fix work?

Bug B — The base is supposed to have a thin **black outline**, but right now the whole base turns solid black.

```
pygame.draw.rect(screen, BLACK, (CENTER_X - 35, HEIGHT - 90, 70, 45))
```

Hint: a missing last number means "fill completely". Add the border thickness.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The ground should stretch **all the way across** the screen. Right now it only covers part of the width.

```
pygame.draw.rect(screen, EARTH, (0, HEIGHT - 50, 200, 50))
```

Hint: the third number is the width of the rectangle.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Recolour the whole scene with your own palette

At the top of the program, the colours are stored as constants. Change `SKY` , `EARTH` , and the greens to your own colours and run it.

Write the colours you chose and what mood they give your scene:

Add one extra detail to the base

Add a small detail to your base — for example an antenna line going up, or a flag rectangle on the side.

Describe the detail you added and the shape you used to draw it:

Hint: an antenna is a `pygame.draw.line` ; a flag is a small `pygame.draw.rect` . Remember to draw it **after** the base so it shows on top.

5 · Make It 📷

Build your own three-part base

Build a scene that has:

1. a ground bar across the bottom,
2. a base body with an outline,
3. a pivot circle on top,
4. one extra detail of your own.

Send a **photo or video** of your base, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I drew the ground by ...

I gave the base an outline by adding ...

The draw order matters because ...

My extra detail is ... and I made it with ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough

Take a video of your three-part base. Talk through it like you are teaching someone. Try to use these words: **ground**, **palette**, **outline**, **draw order**, **pivot**.

1. Show all three parts and name them.
2. Read one rectangle line out loud and explain its four numbers.
3. Explain how draw order decides what sits in front.
4. Show the detail you added on top.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + your photo or video to teacher on KakaoTalk.